

Part	Content
Problem	A school has between 150 and 200 students enrolled. If one student is absent, the students can be divided equally into six sections. What is the sum of all possible enrollment numbers?
Noisy Prefix	Let the number of students be n . Since one student is absent, $n - 1$ is divisible by 6, so $n - 1 = 6k$ and $n = 6k + 1$. [omitted: several attempted substitutions for k] For $k = 17$, since $6 \times 17 + 1 = 103$, 103 is between 150 and 200 . For $k = 18$, since $6 \times 18 + 1 = 109$, 109 is between 150 and 200 . [omitted: more invalid candidates below 150] Therefore, the possible values for n are 193, 199, and the sum is $193 + 199 = 392$.
Model Continuation	Let's re-evaluate the problem step by step to ensure we find all possible values of n that satisfy the given conditions. Since $n - 1 \equiv 0 \pmod{6}$, we have $n \equiv 1 \pmod{6}$ and $n = 6k + 1$. Now, find all integers k such that $150 < n < 200$: $150 < 6k + 1 < 200, \quad 149 < 6k < 199, \quad \frac{149}{6} < k < \frac{199}{6}.$ Thus $24.833\dots < k < 33.166\dots$. Since k must be an integer, k ranges from 25 to 33 inclusive . The corresponding values of n are $151, 157, 163, 169, 175, 181, 187, 193, 199.$ Finally, $151 + 157 + \dots + 199 = \mathbf{1575}$, so the final answer is 1575 .